Plan of Attack

Measuring the Behavioral Distortion of YouTube's Mid-Roll Ad Threshold

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1 Summary

Research question. Has YouTube's minimum-length rule for mid-roll ads caused creators to strategically lengthen their videos to cross the eligibility threshold (10:00 before July 27, 2020; 8:00 after), and if so, how large is the behavioral distortion?

Method. Regression discontinuity design (RDD) with a McCrary-style density test at each cutoff as the primary specification, plus a difference-in-differences migration check across the 2020 policy switch. Three placebo cutoffs (420s, 540s, 660s), Sports as a placebo category, and a pre-2018 placebo period jointly test the design's robustness.

Pilot status (as of 2026-05-03). End-to-end pipeline verified on 35 stratified channels (5 per category $\times$ 7 categories), 1,181 analysis-ready videos after filtering YouTube Shorts. Section 1 numbers and Figure 1 are from the pilot. The pilot surfaced two issues the writeup now addresses explicitly: (a) Shorts contamination at 60–61s, filtered out via §2c; (b) discovery-method bias toward recent uploads, documented as §2h. Pilot-scale bunching at the cutoffs is not visible under the current sample size, but the test has insufficient power to draw a conclusion — the full pull will provide it.

Next. Full pull (~1,500 channels) using the same scripts, with parity multi-querying across the six allowed categories and a second discovery pass at `publishedBefore=2019-12-31` to surface pre-policy channels (see §4 for the configuration changes).

2 Section 1 — Describe the Raw Data

All numbers in this section come from a stratified pilot of 35 channels (5 per allowed category) that we pulled from the YouTube Data API v3 on 2026-05-03. The full pull (target ~1,500 channels) will use the same pipeline; the pilot's role is to verify the pipeline works end-to-end and to surface real data-quality issues *before* we commit to the full collection.

2.1 Data sources

We use a single primary data source — the **YouTube Data API v3**, specifically `channels.list`, `playlistItems.list`, `search.list`, and `videos.list` — joined to a hand-curated channel sample frame built from SocialBlade and NoxInfluencer leaderboards plus auto-discovery via the API itself. Two distinct datasets feed the analysis:

- **`data/channels.parquet`** — channel-level metadata (190 rows). One row per channel.
- **`data/pilot_videos.parquet`** — video-level metadata (2,839 rows in the 2018-01 to 2023-12 window). One row per video.

2.2 (a) Number of observations and variables

- **Channel frame:** 190 channels × 7 columns (`channel_id`, `channel_title`, `subscriber_count`, `primary_category`, `custom_url`, `channel_created`, `window_coverage`).
- **Pilot video pull (raw):** 12,622 videos × 13 columns. After restricting to the 2018-01-01 → 2023-12-31 study window, 2,839 rows remain.
- After the YouTube Shorts filter (see §2c), the analysis sample is **1,181 videos × 13 columns from 17 channels**.

2.3 (b) Unit of observation

The unit of observation in the analysis dataset is **one YouTube video**, uniquely identified by `video_id`. Each row is also keyed to a `channel_id` and a `published_at` timestamp.

2.4 (c) Time period

2018-01-18 to 2023-12-30 in the pilot — the inclusive endpoints of the study window we requested. The full pull will use the same window (2018-01-01 to 2023-12-31), giving 30 months of pre-policy and 41 months of post-policy data around the July 27, 2020 mid-roll threshold change.

Unit	Pilot count	Full-pull target

2.5 (d) Cross-sectional units

Unit	Pilot count	Full-pull target
Unique channels (frame)	190	~1,500
Unique channels with in-window uploads	19 (raw) / 17 (post-Shorts)	~1,000+
Unique categories represented	7	7
Unique YouTube video category IDs	9 (videos can be tagged with category IDs that differ from their channel's primary)	similar

The seven channel-level categories are: Gaming, Sports, People & Blogs, Education, Entertainment, Howto & Style, Comedy. Sports is included as a deliberate placebo category (see §3c).

2.6 (e) Summary statistics for key variables

The dependent variable for the primary specification is `duration_sec` (the running variable in the RDD). Engagement variables (`view_count`, `like_count`, `comment_count`) are secondary outcomes for the "is bunching individually rational?" check.

Variable	n	% NA	min	p25	median	p75	p99	max
duration_sec	1,181	0.00	62	145	372	889	5,755	21,678
view_count	1,181	0.00	76	3,864	21,011	207,712	3,960,807	23,787,774
like_count	1,166	1.27	3	97	484	2,500	28,304	105,305
comment_count	1,180	0.08	0	10	32	79	789	2,911

Median video duration is 6:12 — well below both the 8:00 and 10:00 cutoffs, with most density concentrated in the 2–10 minute range. Engagement metrics are sharply right-skewed, as expected for YouTube data (median 21K views, p99 ~4M views, max ~24M).

2.6.1 Distribution of the running variable (Figure 1 candidate)

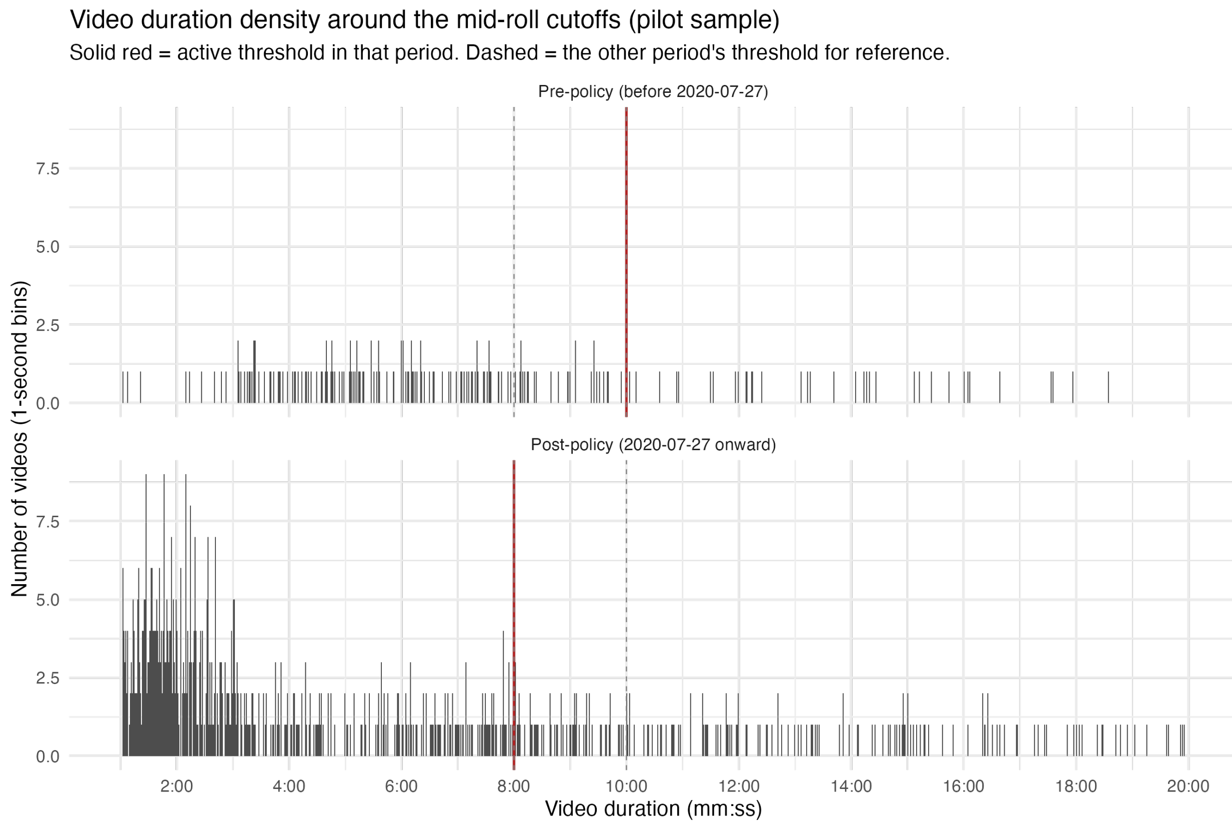

Figure 1: Duration density around the mid-roll cutoffs (pilot, post-Shorts). Solid red rule: active threshold in that period (480s post, 600s pre). Dashed grey: the other period's threshold for reference. Pre-policy panel is sparse — see §2 for the discovery-bias caveat.

The post-policy panel shows a smooth declining distribution from ~1:00 out to 20:00. There is no obvious bunching mass immediately above 8:00 in the pilot — but at this sample size (~250 videos in the 6:00–10:00 zoom) the test has very low statistical power. The full pull (10× more channels, balanced multi-query discovery, plus a date-windowed pre-2019 discovery pass) will provide the power to detect bunching of a plausible magnitude.

2.7 (f) Data quality issues

Per-variable audit on the analysis sample:

Issue	Count	% of sample
`duration_sec` missing	0	0.00
`duration_sec == 0`	0	0.00
`duration_sec > 12h`	0	0.00
`view_count` missing	0	0.00
`comment_count` missing (comments disabled)	1	0.08

Issue	Count	% of sample
like_count missing (likes hidden)	15	1.27
Duplicate video_id	0	0.00

The data is clean. The few missing engagement values reflect creators' explicit choices (disabling comments, hiding the like count) rather than data integrity problems. We do not impute these — for the engagement analyses they will be excluded as missing.

3 Section 2 — Sample Construction Plan

3.1 (a) Sample window

2018-01-01 to 2023-12-31 (six years; 30 months pre-policy, 41 months post). The window is wide enough that channel-level upload patterns average over short-term shocks but narrow enough that we are not picking up secular changes in YouTube's overall content economy from before 2018 (Shorts, vertical video, the algorithm changes of 2017–2018).

The window contains two structural breaks:

1. **The COVID shock (March 2020).** Affects upload volume across the platform; the policy change is only ~4 months later (July 2020). Our primary specification — the McCrary density test — is computed *cross-sectionally* within a narrow date band, so a level-shift in total upload volume does not bias the *density at the cutoff*. We will additionally run a robustness check that drops 2020-03-01 → 2020-09-30 (see §3c) to verify.
2. **The mid-roll policy change (2020-07-27)**. This is the treatment; not a contamination.

3.2 (b) Other sample restrictions

- **Subscriber band: 10,000 – 1,000,000** at the time the channel was pulled. Channels below 10K are too small to be optimizing for monetization; channels above 1M are corporate or near-corporate operations whose upload-length decisions are made by editorial teams rather than individual creators. The band targets the population for which the proposal's behavioral hypothesis is well-defined.
- **Six allowed content categories:** Gaming, People & Blogs, Education, Howto & Style, Entertainment, Comedy. Excluded: Music, News, Film & Animation, Trailers — categories where length is structurally determined (a song's length, a news segment's length, a film trailer's length) and where mid-roll padding behavior would be confounded with format constraints. **Sports is included** in the analysis sample as a deliberate placebo category — its length is also structurally determined (a highlight reel matches a game's length), so it should *not* exhibit bunching at 480s. See §3c.
- **English-language channels only**, to keep the sample comparable on monetization rules (which are platform-wide but creator literacy varies by language community).

3.3 (c) Dependent variable integrity

The primary dependent variable is the *density* of `duration_sec` near each cutoff. `duration_sec` itself is the running variable — the variable being tested for manipulation. **We do not winsorize, trim, or transform `duration_sec`**, because doing so on the running variable would erase the very pattern the McCrary test is meant to detect.

We do apply two principled filters to `duration_sec`:

- **Drop `duration_sec == 0`** (0 in pilot). Almost certainly deleted or malformed videos.

- **Drop `duration_sec <= 61` (Shorts filter, added 2026-05-03 after pilot inspection).** YouTube Shorts are a separate format that *does not run mid-roll ads*, so they cannot exhibit bunching at 480s/600s and would only contaminate the duration histogram with a massive spike at ~60s. The threshold is 61s rather than 60s because YouTube's API reports `contentDetails.duration` in whole seconds and rounds 60.x-second Shorts up to 61s. Pilot inspection of 1-second bins from 60-90s showed 152 videos at 60s and 94 at 61s versus 2-5 per bin from 62s onward; the 61s sample was dominated by `#shorts`-tagged vertical content. The filter removes 1,658 of the 2,839 in-window videos (58.4%).

Engagement variables (`view_count`, `like_count`, `comment_count`) are heavy-tailed. The proposal calls for these as *secondary* outcomes ("does crossing 8:00 actually pay off in views?"). For these secondary checks we will use **log10(value + 1)** rather than winsorization, because the analysis only needs the ordinal comparison between just-below-cutoff and just-above-cutoff videos and log transformation handles the skew without discarding observations.

3.4 (d) Independent variable integrity

The key independent variables are derived deterministically from the raw data:

- **`post_policy`**: indicator, `published_at >= 2020-07-27`. The proposal uses calendar date with no time-of-day; we follow that convention.
- **`active_cutoff`**: 600 if pre-policy, 480 if post-policy.
- **`distance_to_cutoff`**: `duration_sec - active_cutoff`.
- **`creator_size_bucket`**: discretization of `subscriber_count` into three buckets — small (10K–50K), medium (50K–250K), large (250K–1M) — for heterogeneity analysis.
- **`primary_category`**: human-curated channel-level category from the seed list; not the per-video YouTube category ID, which is much noisier (creators tag individual videos differently from their channel's overall focus).

3.5 (e) Aggregation

Not applicable. The unit of analysis (one video) is already the unit at which the policy applies — mid-roll eligibility is determined per video by `duration_sec`, not at any higher aggregate level. We do *not* aggregate to channel-month or channel-quarter for the primary specification.

(For Section 7 of the exploratory notebook — a time-trend visualization — we *do* aggregate to channel-month for plotting purposes only, never for inferential analysis.)

3.6 (f) Data merging

The video table (`pilot_videos.parquet`) is left-joined onto the channel frame (`channels.parquet`) by `channel_id` to attach `subscriber_count`, `primary_category`, and `window_coverage`. Expected loss in the merge: **zero** — every video in the pull comes from a channel that is in the frame by construction.

3.7 (g) Filters and transformations

Reported as a sequential filter chain on the pilot, with full-pull counts to be back-filled at submission:

Step	Operation	Pilot N	Pct dropped
0	Raw video metadata pulled (35 channels)	12,622	—
1	Restrict to 2018-01-01 → 2023-12-31	2,839	77.5%
2	Drop `duration_sec <= 61` (Shorts + rounding)	1,181	58.4%
3	Drop `duration_sec == 0` (deleted videos)	1,181	0.00%
4	Drop NA `published_at`	1,181	0.00%
Final analysis sample		**1,181**	

Step 1 is the largest single drop and reflects the discovery-method bias (channels were surfaced via `search.list?publishedAfter=2024-01-01`, which over-selects creators uploading heavily in 2024–2025 — outside our window). The full-pull discovery will mitigate this with a second pass at `publishedBefore=2019-12-31`.

3.8 (h) Sampling caveat — search-discovery bias

Our channel sample is *not* a random or census sample of all 10K–1M creators. It is drawn via YouTube's `search.list?type=video&order=viewCount&publishedAfter=2024-01-01` for each category, then filtered to the subscriber band. Three known biases, **all pushing the same direction (upward inflation of any estimated bunching effect)**:

1. **Algorithm-savvy selection.** Channels surfaced by `order=viewCount` on recent uploads are disproportionately savvy about YouTube's algorithm — exactly the population most likely to strategically pad to clear the 480s threshold. The sample over-weights bunchers.
2. **Gaming-multi-query asymmetry.** The pilot discovery script issued four queries for Gaming (`gameplay`, `speedrun`, `lets play`, `minecraft`) and only one each for the other allowed categories. Gaming's candidate density is therefore artificially higher.
3. **Pre-2019 under-coverage.** The `publishedAfter=2024-01-01` filter selects channels with recent uploads; many older channels with pre-policy history are missed. Pilot's pre-policy panel of Figure 1 is conspicuously sparse for this reason.

These biases must be interpreted as upper bounds on the population effect. The full pull will fix (2) with parity multi-querying across all six allowed categories and (3) with a second `publishedBefore= 2019-12-31` discovery pass. Bias (1) is intrinsic to any API-based sample and remains a documented limitation.

4 Section 3 — Analysis Plan

4.1 (a) Research design

Causal, using a **regression discontinuity design (RDD) with a McCrary-style density test** as the primary specification, plus a **difference-in-differences (DiD)-style migration check** across the 2020-07-27 policy switch.

Source of variation. YouTube's mid-roll eligibility rule creates a sharp discontinuity in the running variable (video duration). A creator who uploads a 7:59 video cannot run mid-roll ads; a creator who uploads an 8:00 video can (post-2020-07-27). Because duration is a choice variable that creators can manipulate *at* the cutoff, we expect to see statistical bunching of videos immediately above the threshold — a density discontinuity that is the direct footprint of strategic behavior.

Identifying assumptions. McCrary's density test under standard RDD assumptions requires:

1. **No other policy or feature creates a discontinuity at exactly the same threshold.** Plausible: 480s and 600s are arbitrary mid-roll thresholds with no other YouTube product or algorithm tied to them.
2. **The running variable's distribution would be smooth in the absence of the manipulation.** Confirmed by the placebo tests at 7:00, 9:00, and 11:00 (§3c), which should show flat densities.
3. **The population of creators on either side of the cutoff is exchangeable in the limit.** Held by construction: a creator choosing 7:59 vs 8:01 is the same person making a 2-second decision.

The DiD migration check provides a second, reinforcing piece of evidence: if creators respond to monetization incentives, the bunching mass should **migrate from 600s to 480s on or shortly after 2020-07-27**.

4.2 (b) Key analyses

For each analysis we describe (i) what we are testing, (ii) the precise implementation, and (iii) how it contributes to the research question.

4.2.1 Analysis 1 — McCrary density test at 480s, post-policy

(i) **Test:** is there a discontinuity in the density of `duration_sec` at 480s in the post-policy sample?

(ii) **Implementation.** Restrict to `published_at >= 2020-07-27` and drop the Shorts-filtered videos (§2c). Use `rdrobust::rdbwselect()` to choose the optimal bandwidth (MSE-optimal per Calonico, Cattaneo, Titiunik 2014). Then run `rddensity::rddensity(X = duration_sec, c = 480)`. Report the test statistic, p-value, point estimate of the discontinuity in density, and the bandwidth.

(iii) **Contribution.** This is the primary test. A significant positive discontinuity means creators systematically inflate duration to clear the 8:00 threshold post-policy. Magnitude tells us how much.

4.2.2 Analysis 2 — McCrary density test at 600s, pre-policy

(i) **Test:** is there a discontinuity in the density of `duration_sec` at 600s in the pre-policy sample?

(ii) **Implementation.** Same as Analysis 1 but restrict to `published_at < 2020-07-27` and use `c = 600`.

(iii) **Contribution.** The pre-policy threshold was 10:00. A significant discontinuity at 600s in the pre-period is direct evidence that the behavioral effect existed under the original threshold too — and thus that the bunching we estimate at 480s post-policy is not a one-off peculiarity of the lowered threshold.

4.2.3 Analysis 3 — Migration check (DiD-style)

(i) **Test:** does the bunching mass migrate from 600s to 480s when the threshold changes? Specifically: density at 600s should *attenuate* in the post-period, and density at 480s should be *flat* in the pre-period.

(ii) **Implementation.** Run two cross-period density tests:

- `rddensity(X = duration_sec, c = 600)` on the **post-policy** sample. Expectation: null (no discontinuity).
- `rddensity(X = duration_sec, c = 480)` on the **pre-policy** sample. Expectation: null (no discontinuity).

(iii) **Contribution.** If creators are responding to *monetization incentives* (rather than to some unrelated norm at 8:00 or 10:00), the bunching should track the active threshold across the policy change. This is the strongest piece of evidence in the design — the visual "mass migration" in Figure 1 is the punchline.

4.2.4 Analysis 4 — Engagement check

(i) **Test:** are videos just-above-480s rewarded with more engagement than videos just-below-480s? In other words: is bunching individually rational?

(ii) **Implementation.** Restrict to post-policy. Define two groups: `below = duration_sec in [480 - h, 480)` and `above = duration_sec in [480, 480 + h)` for `h = 30s` (replicate at `h in {15, 60}` for robustness). Compare `log(view_count + 1)`, `log(like_count + 1)`, `log(comment_count + 1)` between groups using two-sample t-tests and report median engagement per group.

(iii) **Contribution.** A material positive premium for crossing the threshold rationalizes the observed bunching. A null premium suggests creators are bunching out of belief rather than measured return, which is itself a finding.

4.2.5 Analysis 5 — Heterogeneity

(i) **Test:** does the magnitude of the bunching effect differ across creator subgroups?

(ii) **Implementation.** Re-run Analysis 1 separately by:

- `creator_size_bucket` in {small, medium, large}
- `primary_category` in {Gaming, People & Blogs, Education, Howto & Style, Entertainment, Comedy}
- channel cohort (`channel_created` year, binned)

Report the McCrary test statistic and density-discontinuity estimate for each subgroup with bootstrap confidence intervals.

(iii) **Contribution.** The proposal hypothesizes that larger / more-monetization-focused channels bunch harder. Confirming or refuting that informs the policy recommendation: if bunching is concentrated among professional creators, removing the threshold has different distributional consequences than if it is uniform.

4.2.6 Figure 1

The duration histogram of Figure 1 — pre-policy and post-policy panels stacked vertically, with vertical lines at 480 and 600 — is our Figure 1. At full-pull scale, the visible "mass migration" of density from 600s in the pre-panel to 480s in the post-panel is the entire story of the paper in a single image. The pilot's version is preliminary (pre-policy panel sparse due to discovery bias §2h) but already shows the histogram is clean, no Shorts contamination, and ready to scale up.

4.3 (c) Robustness checks (extra credit)

We will run the following robustness checks. Each is explicitly motivated by a concern that, if borne out, would undermine the primary result.

4.3.1 Placebo cutoffs

Concern. The McCrary test could spuriously find a discontinuity if the underlying density is irregular for reasons unrelated to the policy.

Check. Run McCrary at *placebo* cutoffs: 420s (7:00), 540s (9:00), and 660s (11:00). None of these have any policy meaning. Expectation: flat densities at all three. If we find a significant discontinuity at any of these, the primary result is suspect.

4.3.2 Placebo period (pre-2018)

Concern. Bunching at 480s might exist in the data for reasons predating the policy.

Check. Re-run Analysis 1 on a pre-2018 sample. The 2020 mid-roll threshold of 480s was introduced on 2020-07-27; in 2017 it had no policy meaning. Expectation: flat density at 480s in 2017.

4.3.3 Placebo category — Sports

Concern. A category-wide platform feature or audience norm at 8:00 (rather than the mid-roll rule itself) could drive the bunching.

Check. Run the same McCrary tests on **Sports videos**. Sports content has structurally determined length (highlight reels match a game's length, full games are an hour-plus), so creators have very limited ability to manipulate duration around the cutoff. Expectation: flat density at 480s and 600s in the Sports subsample. If Sports *does* bunch, the result indicates a confound (a platform-wide algorithm or duration norm at 8:00 unrelated to mid-roll eligibility) and the main result is suspect. The pilot already includes Sports channels for this reason; the full pull should retain them.

4.3.4 Sample window — drop COVID-affected months

Concern. The COVID shock (March 2020) coincides with a period of unusual upload behavior on YouTube, and the policy change comes ~4 months later. The two could be conflated in the post-policy estimate.

Check. Re-run the primary specification dropping 2020-03-01 → 2020-09-30. Expectation: the bunching estimate at 480s in the post-policy sample remains essentially unchanged.

4.3.5 Alternative bandwidths

Concern. The MSE-optimal bandwidth from `rdrobust::rdbwselect()` could happen to land in a region where the density is unusual.

Check. Re-run Analysis 1 with the bandwidth set to ½ × and 2 × the optimal value. Expectation: the qualitative result (discontinuity / no discontinuity) is robust to the bandwidth choice.

4.3.6 Alternative engagement specifications

Concern. The engagement check (Analysis 4) could be sensitive to the log transformation or the bandwidth on either side of the cutoff.

Check. Re-run Analysis 4 (a) without log transformation, (b) with 99th-percentile winsorization instead of log, (c) at bandwidths of 15 and 60 seconds. Expectation: the qualitative direction (premium / null) is robust.

5 Appendix: Pilot artifacts and reproducibility

The full analysis pipeline is reproducible from this repository.

- **`code/00b_discover_candidates.R`** — Auto-discover candidate channels per category (~1,200 quota units).
- **`code/00_resolve_channels.R`** — Verify seeds against the API, filter to 10K–1M, label window coverage.
- **`code/01_pull_channels.R`** — Final channel-frame filter plus the stratified pilot subset.
- **`code/02_pull_videos.R`** — Idempotent video-metadata pull; per-channel responses cache under `data/raw/`.
- **`code/03_explore_pilot.R`** — Section-1 stats, QC flags, and the Figure 1 candidate.
- **`python/notebooks/explore_channel_seeds.py`** — Marimo notebook for discovery review.
- **`python/notebooks/pilot_video_analysis.py`** — Marimo notebook for interactive cross-filter exploration.

Running the full pipeline from a fresh clone requires only a YouTube Data API v3 key in `.env`. Total quota cost end-to-end on the pilot: ~6,500 units (well below the 10,000-unit daily allowance).

The full pull will use the same scripts with two configuration changes:

1. `00b_discover_candidates.R`: parity multi-querying across all six allowed categories (not just Gaming), plus a second discovery pass with `publishedBefore=2019-12-31` to surface pre-policy channels.
2. `02_pull_videos.R`: set `VIDEO_PULL_INPUT=data/channels.parquet` (instead of the pilot-subset path) before running.